

Seismic Wave Propagation

Learning Objectives

Knowledge Level:

- Wave propagation principles
- P and S waves
- Snell's law
- Converted waves



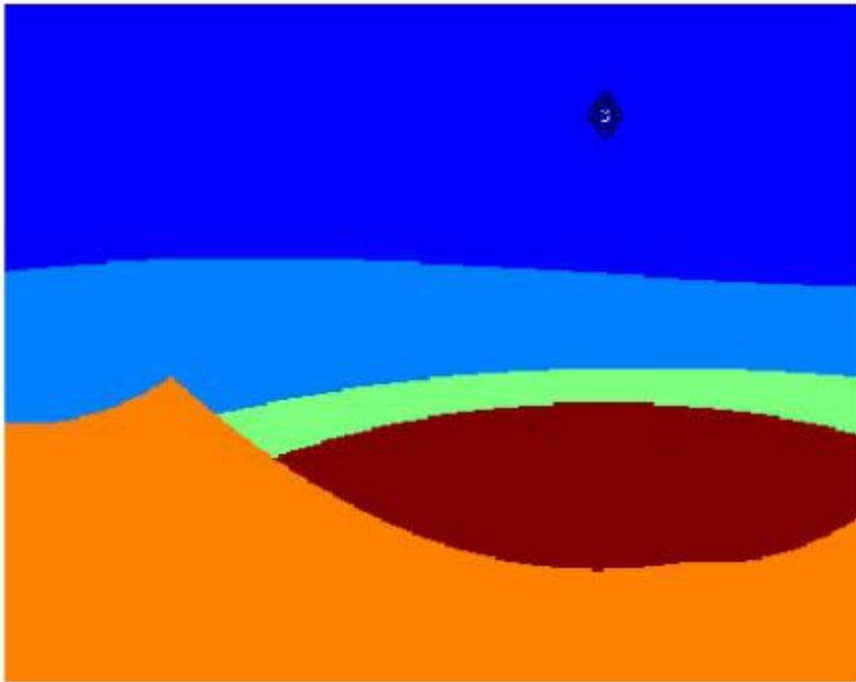
TOPICS

- Seismic Wave Propagation
- Rays, Wavefronts, and Diffractions
- Earth Velocities
- Snell's Law
- Converted Waves

WHY IMPORTANT

- Seismic energy and noise propagate as waves
- Velocity of propagation controls arrival times and patterns of arrivals at surfaces
- Seismic waves require geometric description
- Wavefront/ray bending affects processing and acquisition
- Diffractions and converted waves contain seismic signal information
- Seismic energy propagates as elastic or stress waves
- Velocity of propagation depends on the elastic properties of rock

Understanding Wave Behavior



DEFINITIONS

ELASTIC - A solid in which **stress** is proportional to **strain**.

STRESS - Force applied perpendicular or parallel to a surface of a body divided by the area of that surface.

STRAIN - Ratio of change in dimension(s), caused by an applied stress, to the original dimension(s).

DEFINITIONS(continued)

HOMOGENEOUS - A homogeneous body has uniform properties throughout.

INHOMOGENEOUS – Properties change either continuously or discretely

ISOTROPIC - In an isotropic body elastic parameters are independent of direction. The seismic propagation velocity in an isotropic medium is the same in all directions of propagation.

ANISOTROPIC - Variation of the elastic parameters (e.g. velocity) depending on the direction of propagation.

The subsurface in general is inhomogeneous and anisotropic. Much in acquisition and processing assumes homogeneous and isotropic media.

Wave Equation

- An equation which relates the spatial and time dependence of a disturbance propagating as a wave. In rectangular coordinates x,y, z it is

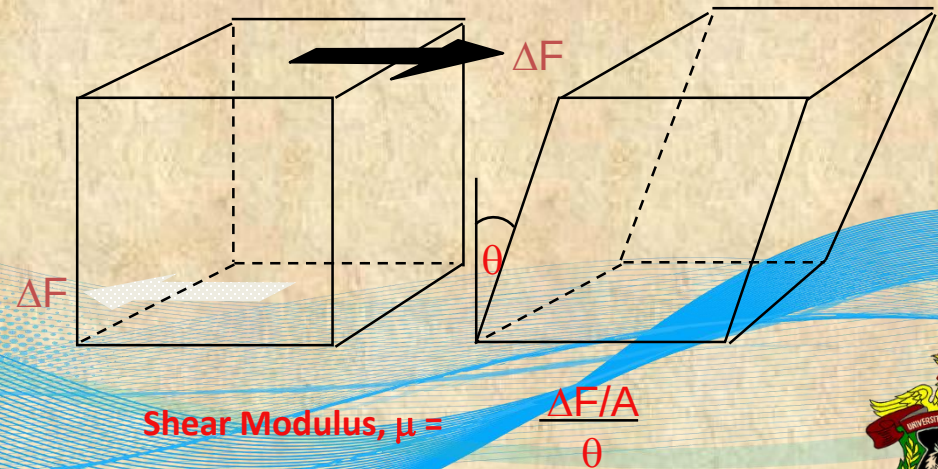
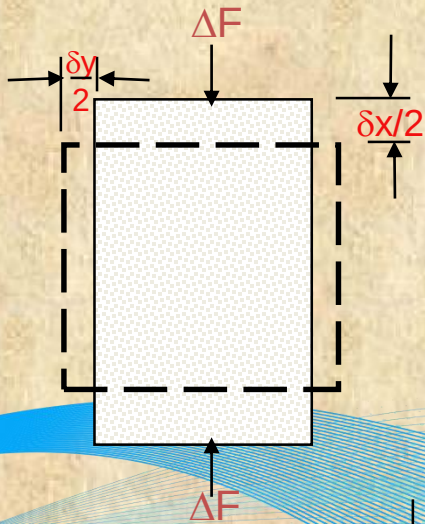
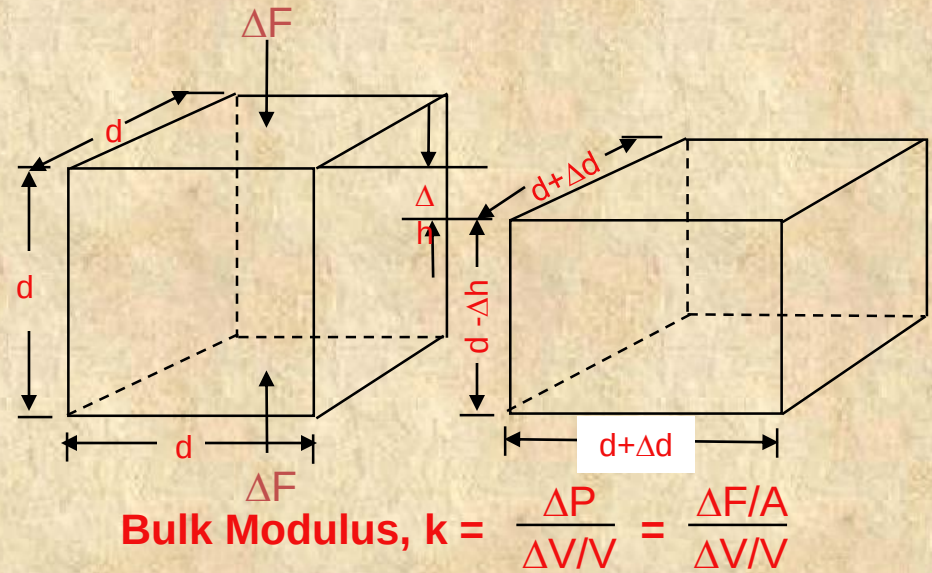
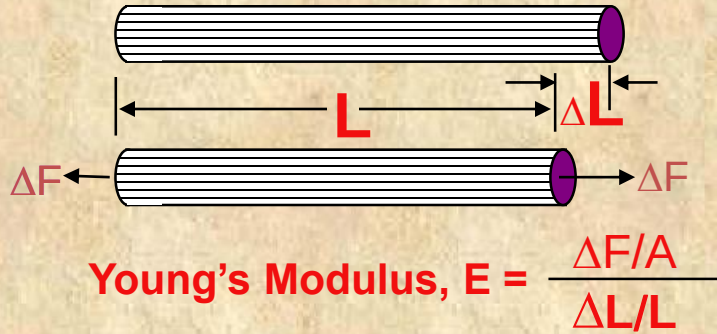
$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \left(\frac{1}{v^2} \right) \frac{\partial^2 \psi}{\partial t^2}$$

Where ψ represents wave displacement (Pressure, rotation, dilatation, etc)

V the velocity of the wave

- In seismic exploration we utilize primary (p) waves which are governed by a scalar wave equation and sometimes shear (s) waves which require more complicated vector treatment.

Elastic Moduli



Seismic Waves Velocities

- In seismic exploration the two waves speeds are the compressional wave velocity (V_p) and shear wave velocity (V_s)
- The wave velocities are the function of three parameters, λ , μ and ρ

And

Where λ is one of the Lamé's coefficients, μ is the rigidity modulus and ρ is the density. Both relationships can be expressed in terms of Young's modulus (E) and Poisson's ratio (σ)

$$\mu = \frac{E}{2(1 + \sigma)}$$

Note that σ (which is sensitive to pore fluid properties) can be expressed in terms of V_p and V_s .

$$\sigma = \frac{\frac{V_p^2}{2V_s^2} - 1}{\frac{V_p^2}{V_s^2} - 1}$$

Seismic Wave Types

Primary (P)

- Fastest
- “Acoustic” or scalar wave equation
- Velocity or pressure detectors

Shear (S)

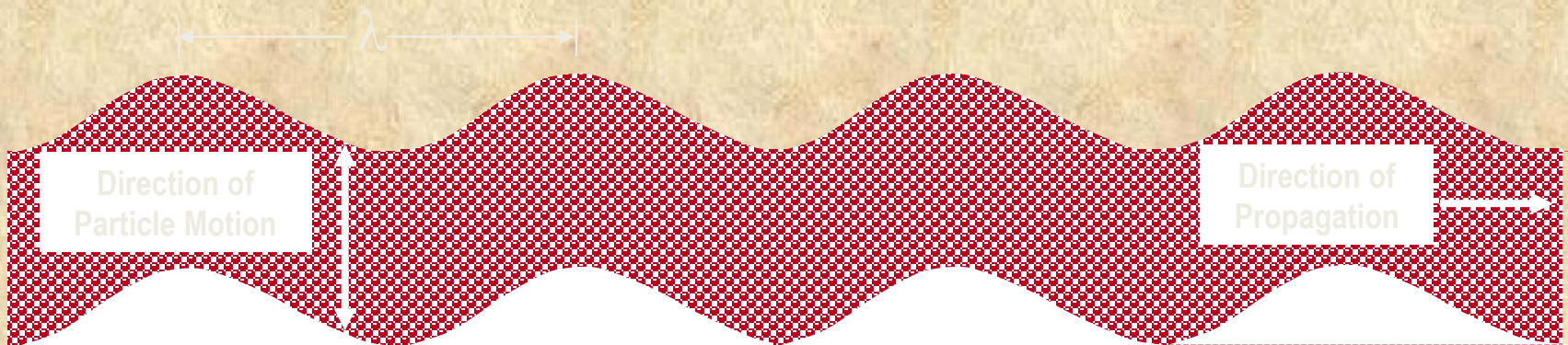
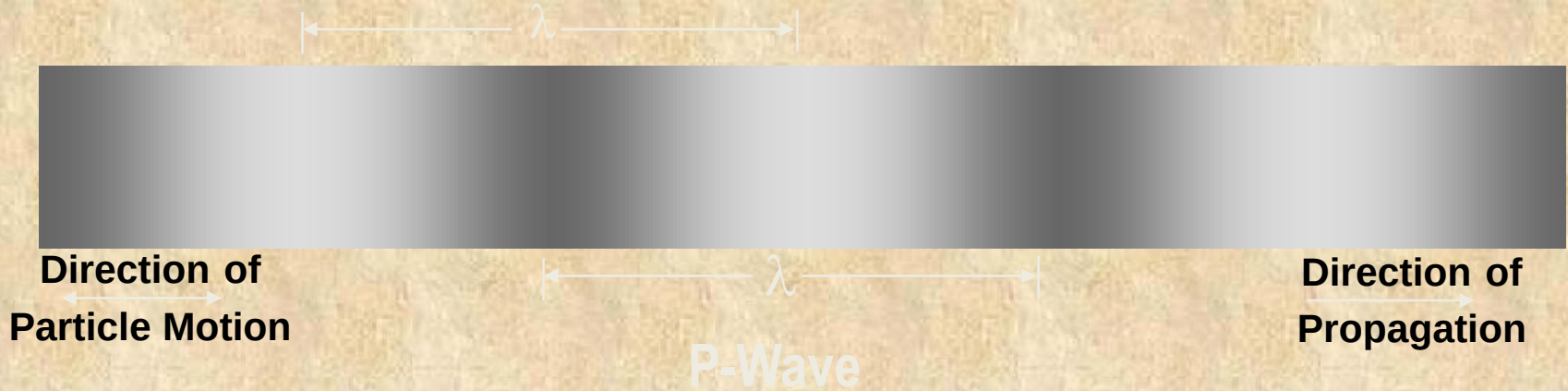
- Slower
- Vector wave equation
- Polarized (SH, SV)
- Horizontal detectors

Surface or Guided

- Rayleigh
- Love
- Dispersive
- Noise (usually low frequency)

ALL WAVES ARE 3D

P- and S-Waves



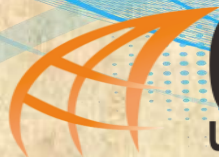
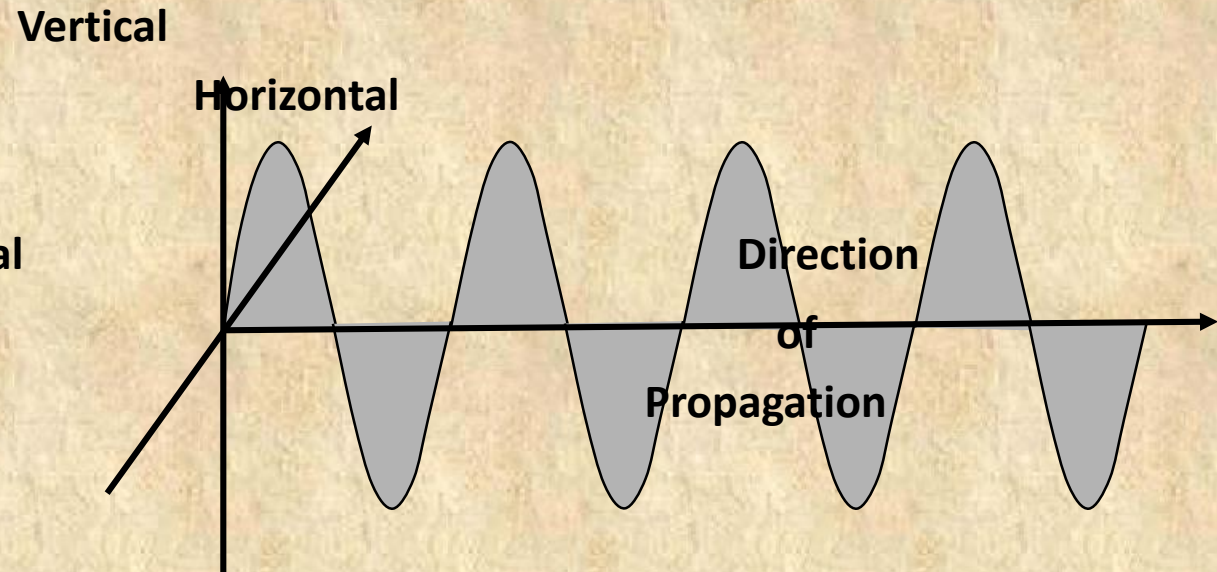
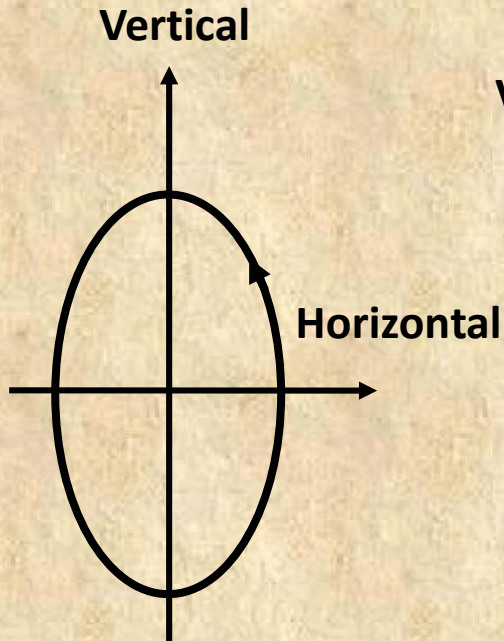
S-Wave



Rayleigh Waves

Waves guided by earth's surface

Particle motion is retrograde ellipse

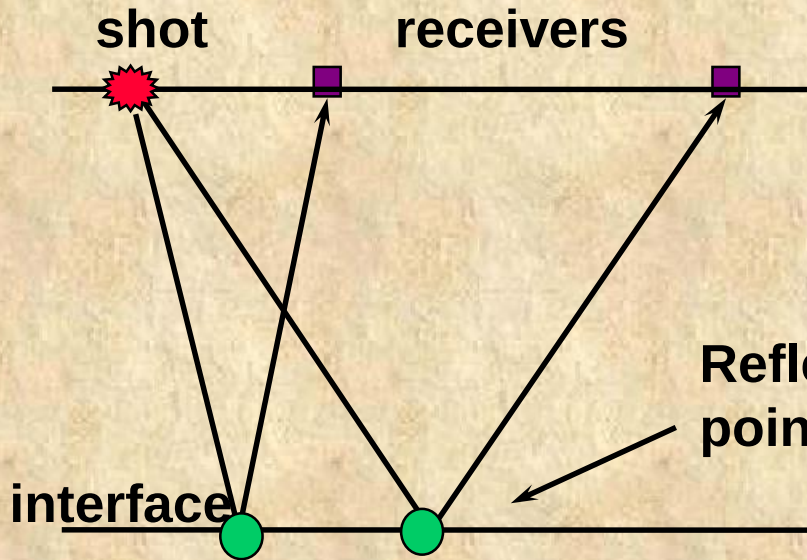


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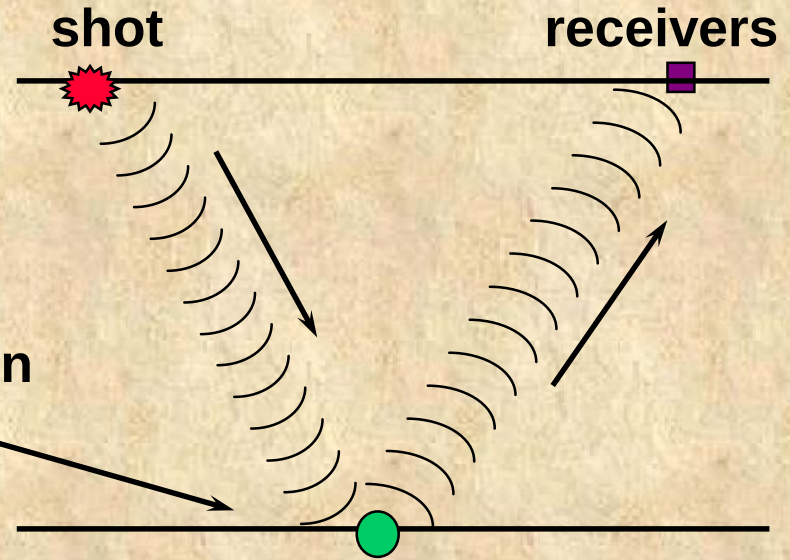


Raypaths and Wavefronts



Raypaths

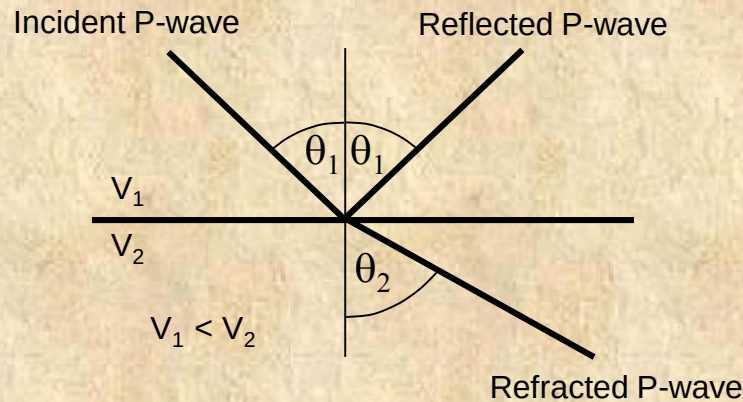
- Rays perpendicular to wavefronts
- Simple to use
- Ray trace modelling



Wavefronts

- How waves actually travel
- Surface of equal travel time
- Surface of equal phase

Snell's Law Of Reflection And Refraction



Snell's Law is a simple model for the behavior of seismic energy obliquely incident upon the boundary between two layers of different velocities.

The incident energy is split into two parts: one reflected at an angle equal to the angle of incidence and one part refracted to an angle related to the velocity contrast between the two layers.

Snell's Law of Reflection:

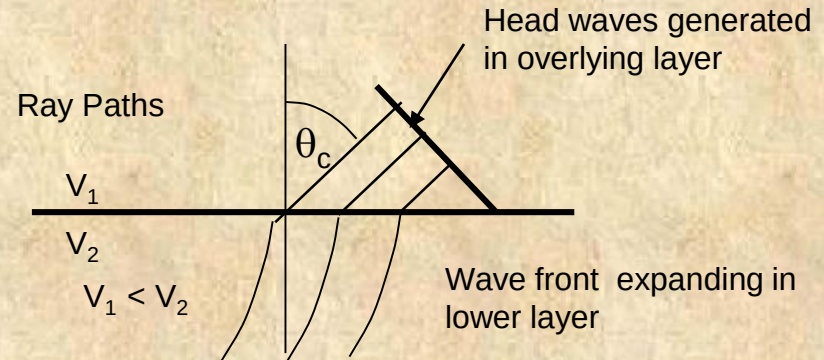
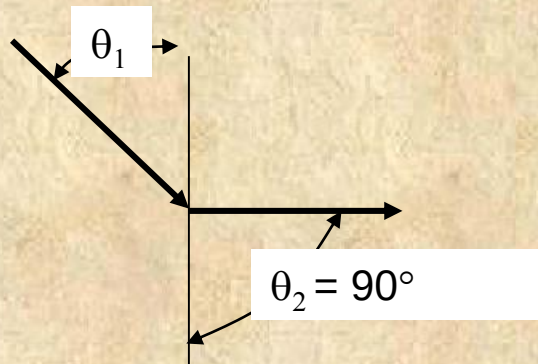
Angle of incidence = angle of reflection

Snell's Law of Refraction:

$$\sin\theta_1/V_1 = \sin\theta_2/V_2$$

$$\sin\theta_1/\sin\theta_2 = V_1/V_2$$

Snell's Law Of Reflection And Refraction



A special case of refraction occurs when the angle θ_2 equals 90° .

Refracted energy propagates horizontally along the boundary at the velocity of the lower layer.

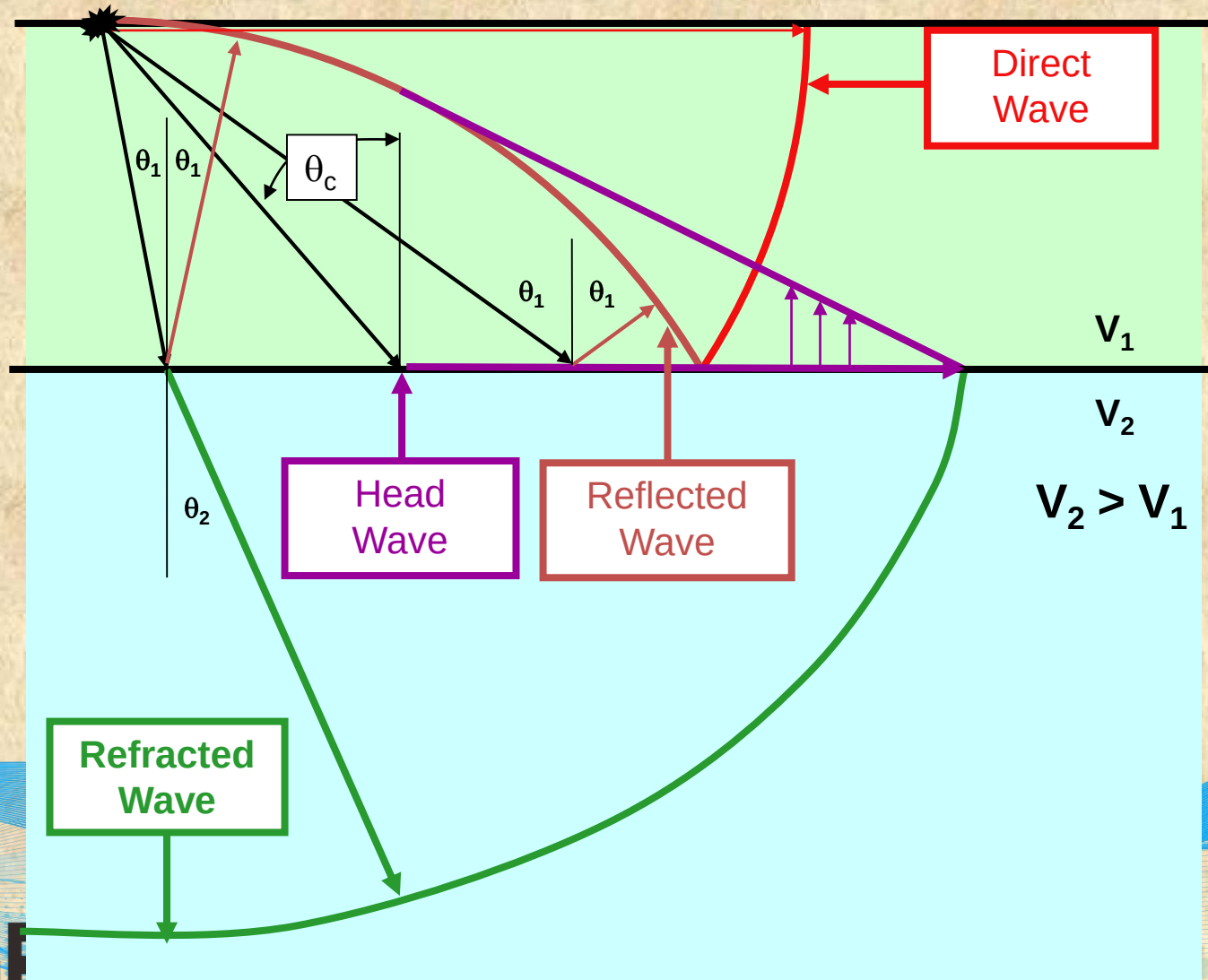
This horizontally traveling energy is called the head wave. It is useful in refraction seismic methods and when processing land seismic data (refraction statics calculation).

The incident angle at which the head wave is first produced is called the “critical angle” and can be derived from Snell’s Law.

$$\theta_c = \sin^{-1}(V_1 / V_2)$$

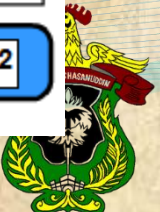
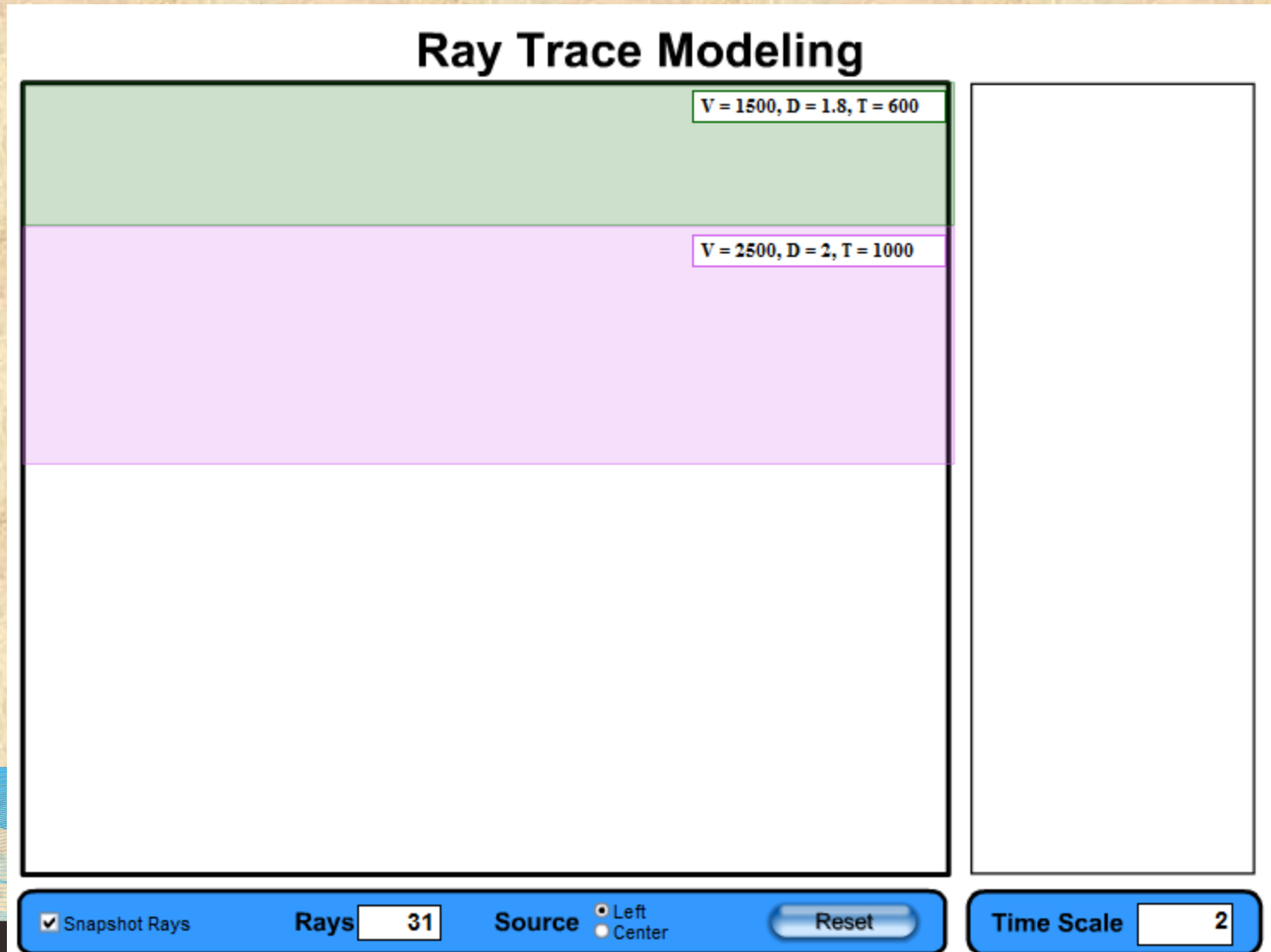
Energy arriving at angles greater than the critical angle is not refracted into lower layers.

Reflected and Refracted Waves



Reflected and Refracted Waves

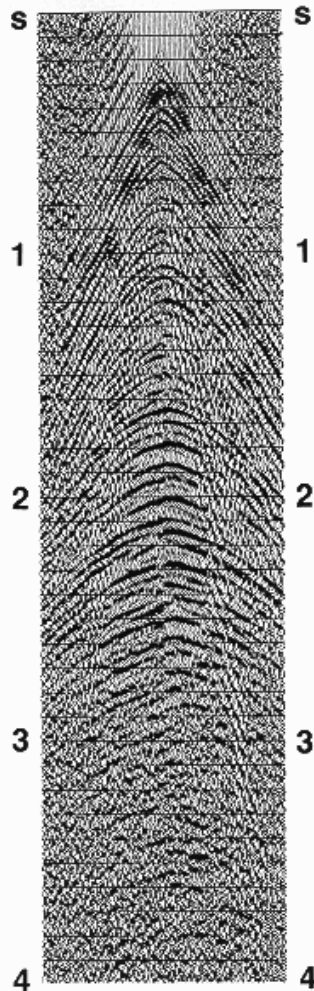
Click in layer
to turn multiples
and head waves
On or off.



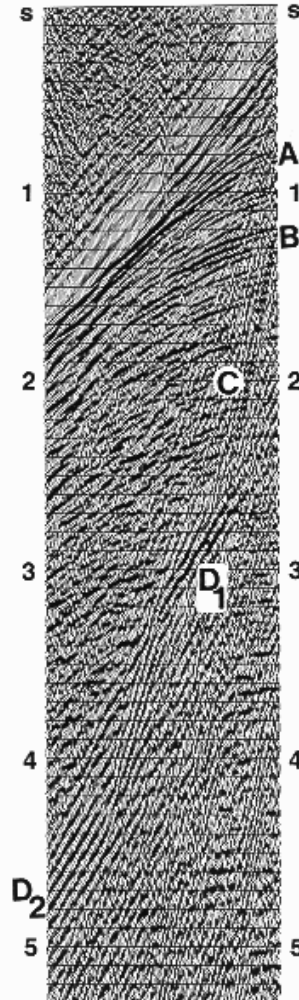
Seismic Field Records

Common Shot Records

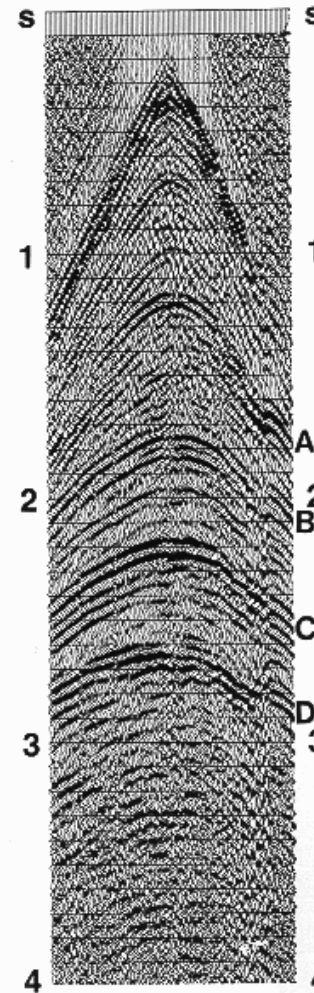
Alaska V



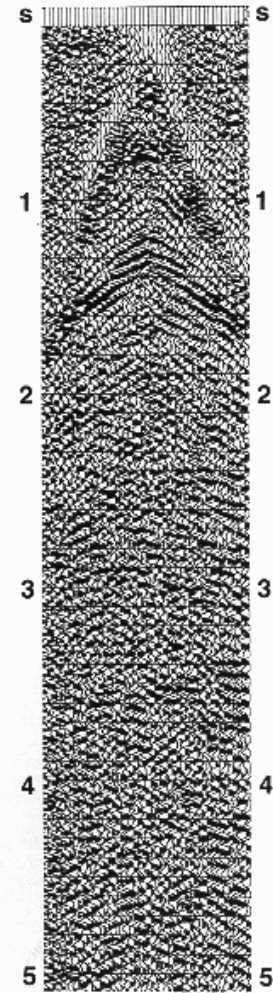
North Africa V



Alaska V

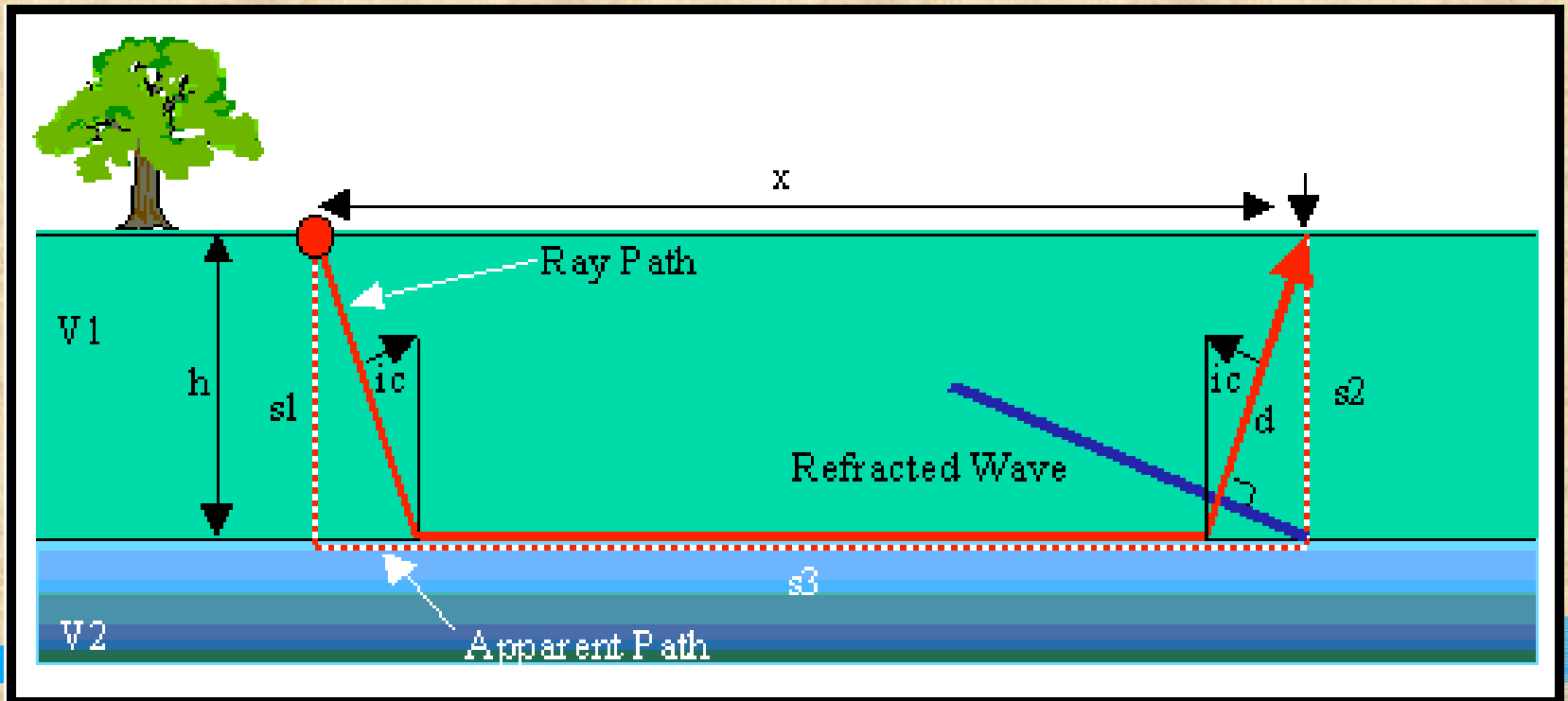


Mississippi V



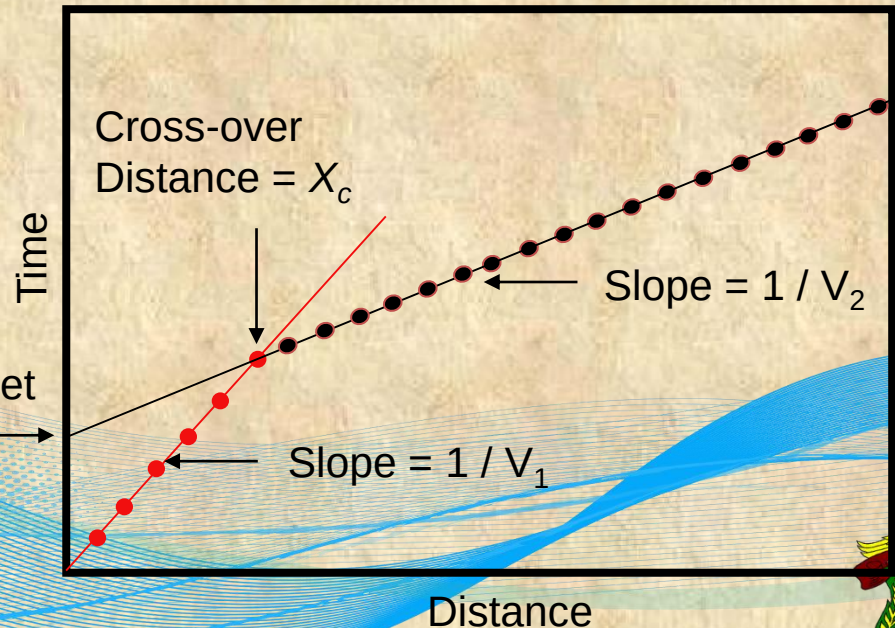
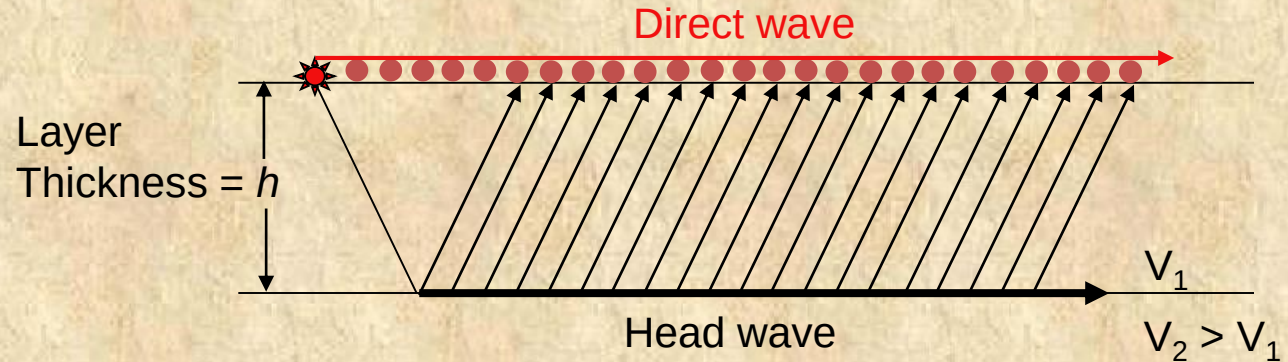
Refraction Seismic Method

- Basis for refraction statics



Refraction Seismic Method

- Seismic source into array of receivers



$$h = \frac{t_0 V_2 V_1}{2(V_2^2 - V_1^2)^{1/2}}$$

or

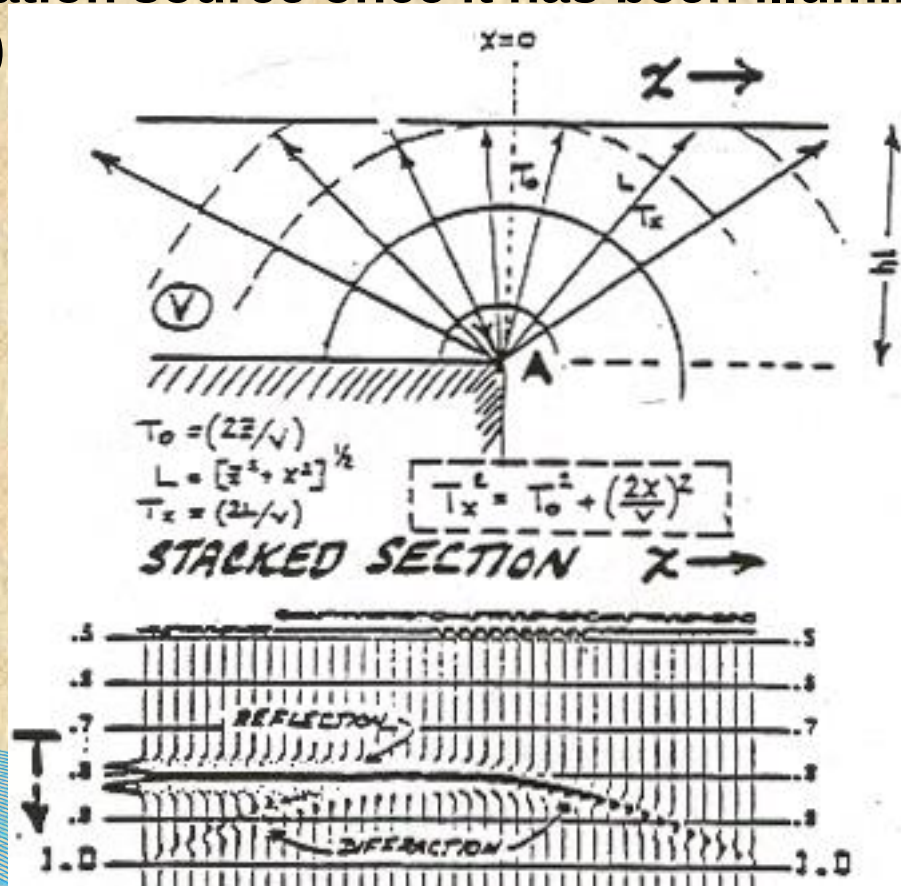
$$h = \frac{x_c}{2} \left(\frac{V_2 - V_1}{V_2 + V_1} \right)^{1/2}$$

KA



Diffractions: Geometry

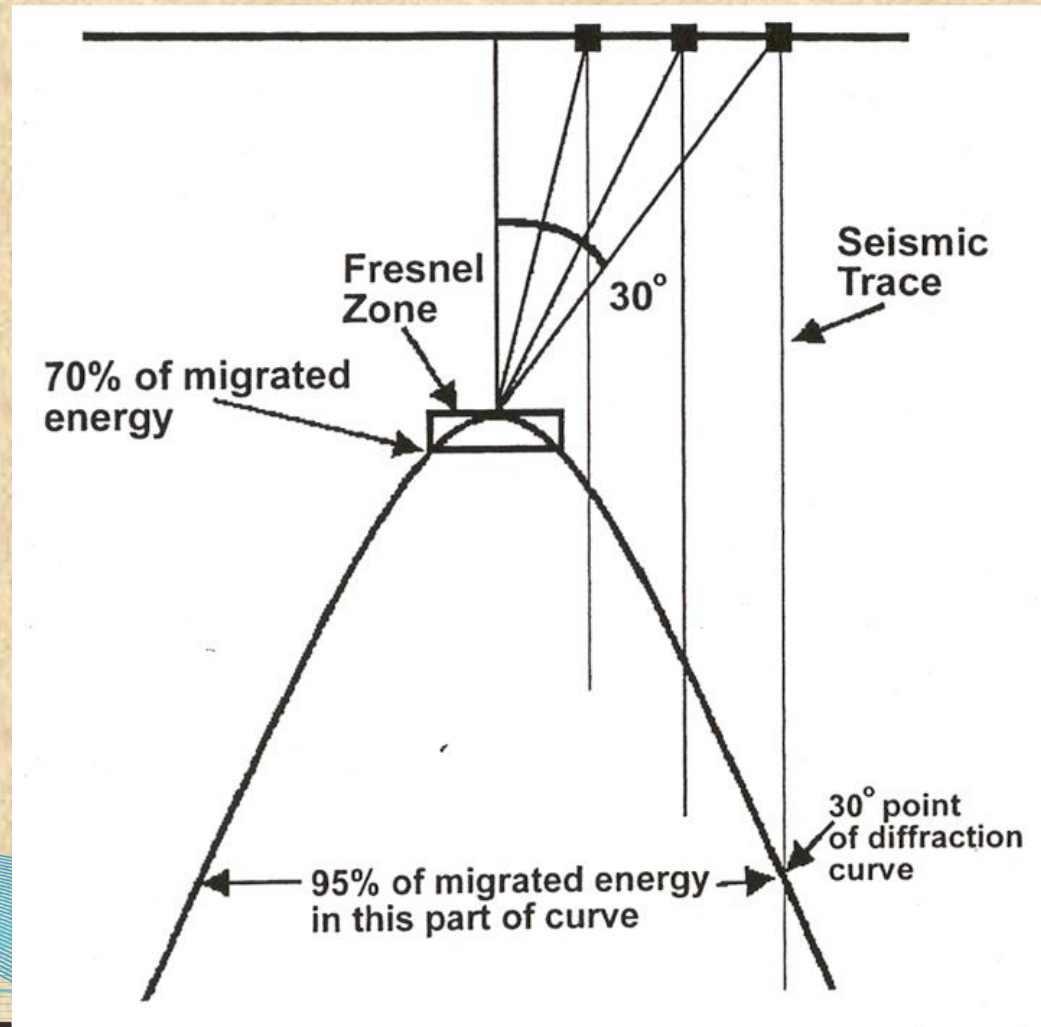
Any sharp bend in a reflecting interface will produce an observable diffraction pattern. The point (A), below, acts like a radiation source once it has been illuminated by source (S)



Diffraction Comments

- Seismic wavefronts are treated theoretically as superposition of subsurface diffractions
- Diffractions are response of point scatterers in subsurface
- Accurate diffraction acquisition and processing are key to imaging
- Design of acquisition must include capture of diffraction arrivals

Diffraction Anatomy



Huygen's Principle

Drag earth
surface downward
and view Seismic
to see effects of
downward continuation

The screenshot shows a software interface for a seismic simulation. At the top, a blue double-headed arrow labeled "Survey Extent" spans the width of the main display area. Below this, a horizontal line of red triangles points downwards. The main display area is a large white rectangle. At the bottom of the interface, there is a control panel. On the left, a "Seismic" button is visible. To its right is a row of ten colored squares, each labeled with a percentage range: 0-10% (grey), 10-20% (purple), 20-30% (blue), 30-40% (dark blue), 40-50% (light blue), 50-60% (green), 60-70% (light green), 70-80% (yellow), 80-90% (orange), and 90-100% (red). Below these are radio buttons for "Model", "Migrate", and "Pause", followed by a "Reset" button. A vertical slider labeled "Velocity" is also present. The bottom right section, titled "Huygen's Principle", contains a text box explaining the principle: "Every point on a propagating wavefront serves as the source of spherical secondary waves, such that the wavefront at some later time is the envelope of these secondary waves."

Survey Extent

Seismic

0-10% 10-20% 20-30% 30-40% 40-50% 50-60% 60-70% 70-80% 80-90% 90-100%

☐ Model ☐ Migrate ☐ Pause

Exploding Reflector Models

☐ Point Diffractor ☐ Syncline ☐ Anticline
☐ Flat Layer ☐ Monocline
☐ Dipping Layer ☐ Fault

Velocity

Huygen's Principle

Every point on a propagating wavefront serves as the source of spherical secondary waves, such that the wavefront at some later time is the envelope of these secondary waves.



Velocity of seismic wave propagation is dependent on subsurface geological and petrophysical properties.



Velocity Definitions

- **Instantaneous velocity**
 - Velocity at which a seismic wave propagates at a point within the medium
 - Varies with wave type or propagation mode
 - P-wave velocity, V_p , is always the fastest
 - S-wave velocity, V_s , is the second fastest
 - V_p/V_s greater than or equal to 2
 - Rayleigh wave velocity is usually about 92% of S-wave velocity
 - Love wave velocity is slightly slower than Rayleigh wave velocity
 - Depends on rock type, age, depth of burial, porosity, pore shape, pore fluid, confining pressure, etc

Velocity Definitions

- **Interval velocity** (V_i) is the average propagation velocity through a depth or time interval = thickness of depth interval divided by vertical time through the interval = $\Delta z_i / \Delta t_i$
- **Average velocity** (V) is total depth to a reflector (Z_r) divided by time to the reflector or twice the depth to the reflector divided by two-way, zero-offset, reflection time = $2Z_r / T_r$
 - Calculated from interval velocities by:

$$V_n = \frac{\sum_{i=1}^n V_i \Delta t_i}{\sum_{i=1}^n \Delta t_i} = \frac{\sum_{i=1}^n V_i \Delta t_i}{T_r}$$



Velocity Definitions

- **Root Mean Square (RMS) velocity $\tilde{V}$ (V)** is the square root of the average squared velocity and is calculated from:

$$\tilde{V}_n = \sqrt{\frac{\sum_{i=1}^n V_i^2 \Delta t_i}{\sum_{i=1}^n \Delta t_i}}$$

where V_i = interval velocity of the i^{th} depth or time interval

Δt_i = two-way time of i^{th} time interval or through the i^{th} depth interval

Velocity Definitions

- **NMO velocity** is the velocity used to correct for Normal Moveout (NMO), i.e. to make primary reflections on CMP gather records occur at the same time on all traces
 - For plane horizontal or gently dipping layers, NMO velocity = RMS velocity

$$T^2 = T_o^2 + \frac{X^2}{V_{NMO}^2}$$

- **Stacking velocity** is the velocity that gives the optimum CMP stack output when used for NMO corrections

Velocity Definitions

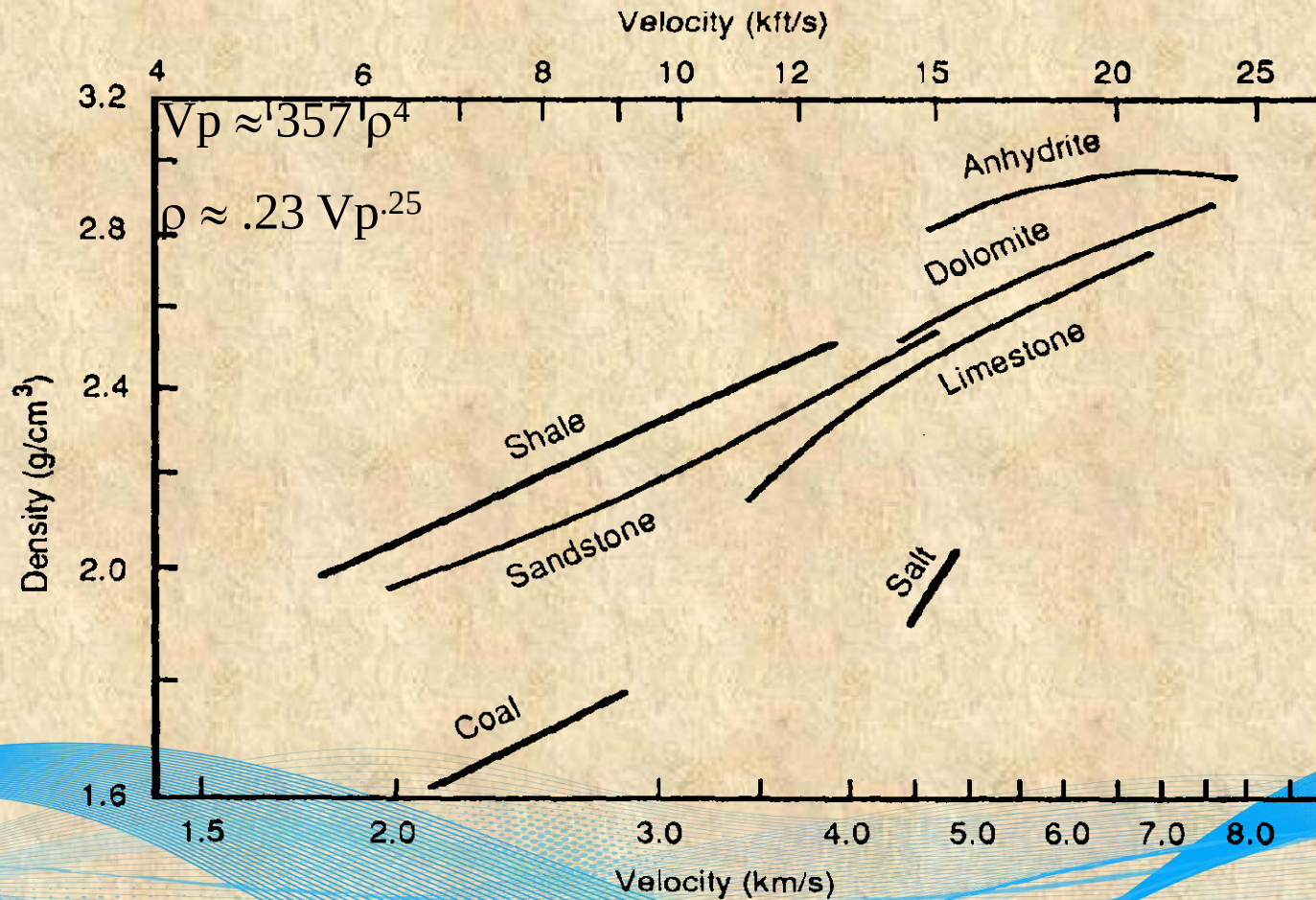
- **Migration velocity** is that velocity which optimizes the output of a migration algorithm, i.e. - repositioning of the reflected energy to the correct locations
- **Apparent velocity** is the velocity at which seismic waves appear to propagate along the surface of the earth

Velocity Model

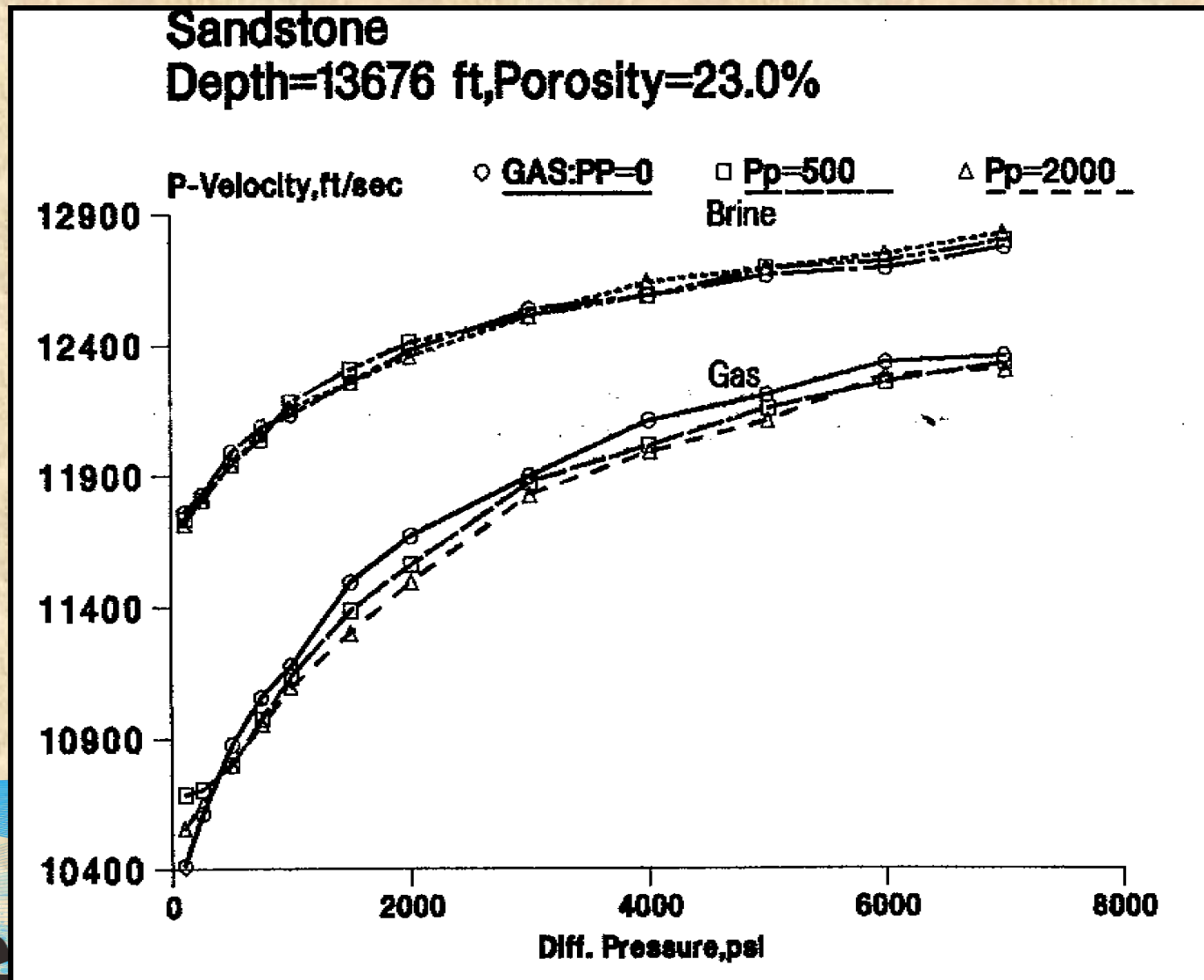
	Layer Thickness (m)	V_{INT} (m/s)	V_{RMS} (m/s)	V_{AVG} (m/s)
1	200	2000	2000	2000
2	300	3000	2550	2500
3	400	4000	3800	3000
4	350	3500	4540	3125
5	500	5000		

Note that RMS velocity increases even when interval velocity decreases.
Also note that $V_{RMS} > V_{AVG}$ below layer1.

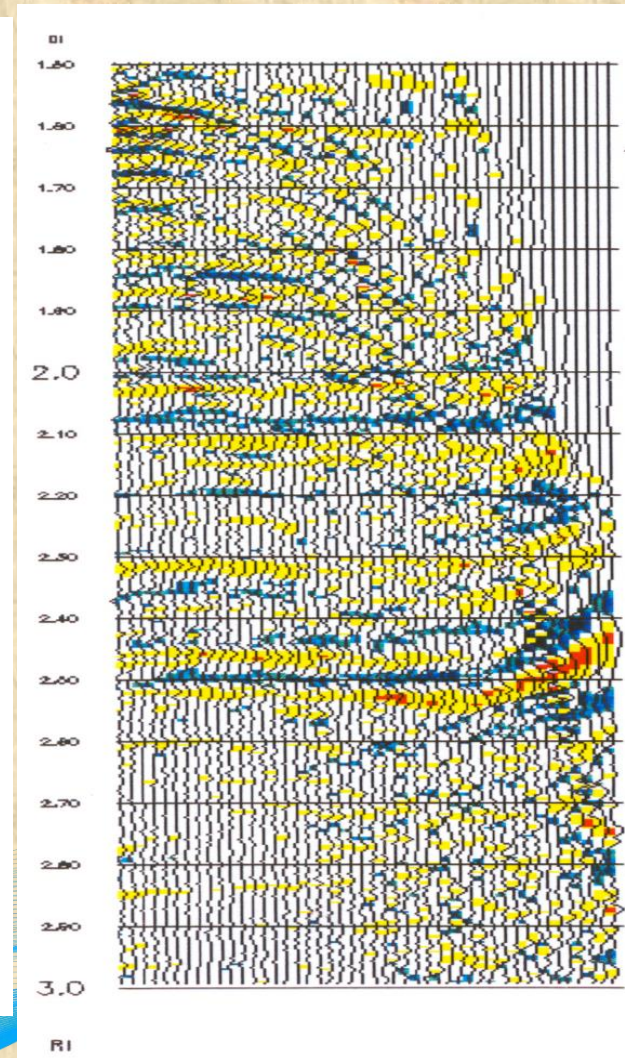
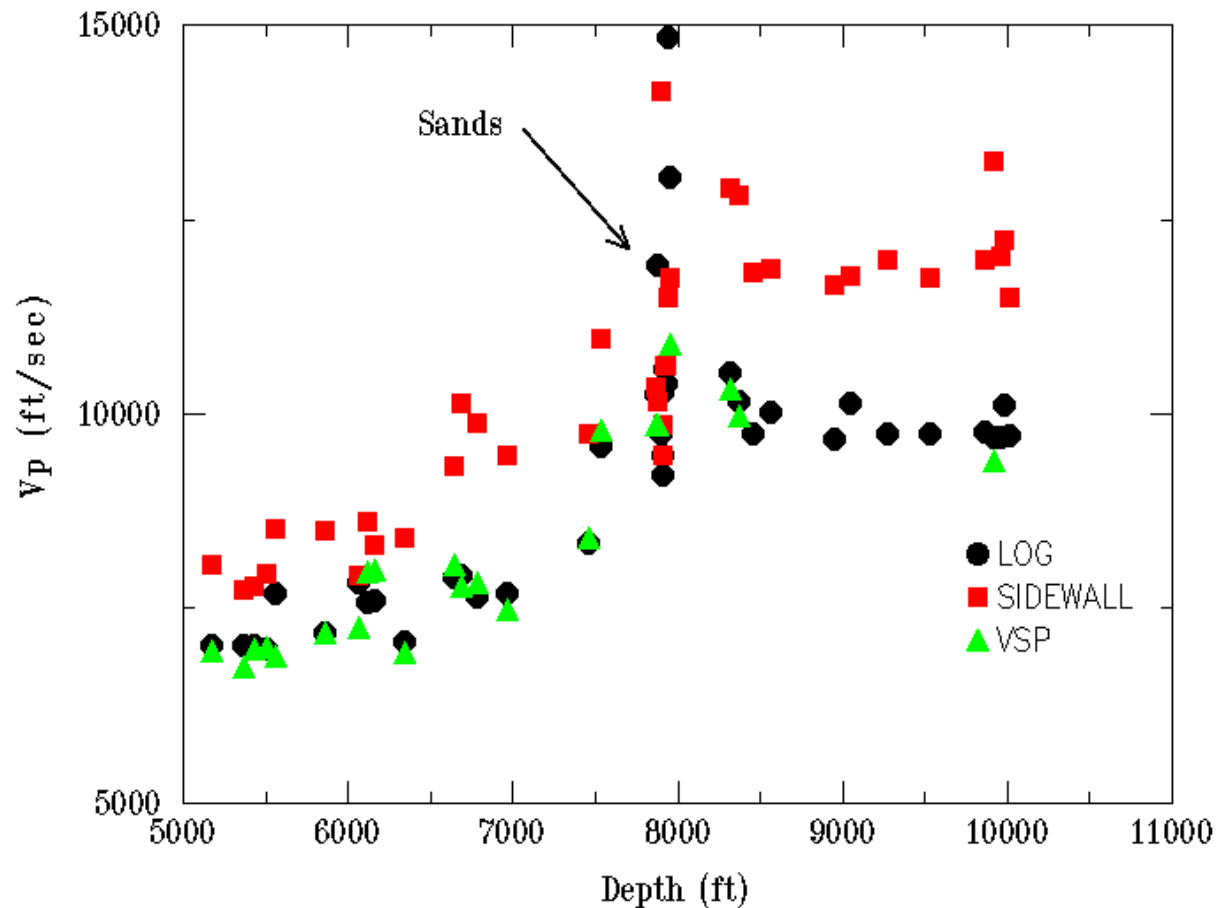
Variation of Velocity and Density



Velocity Dependence on Effective Pressure



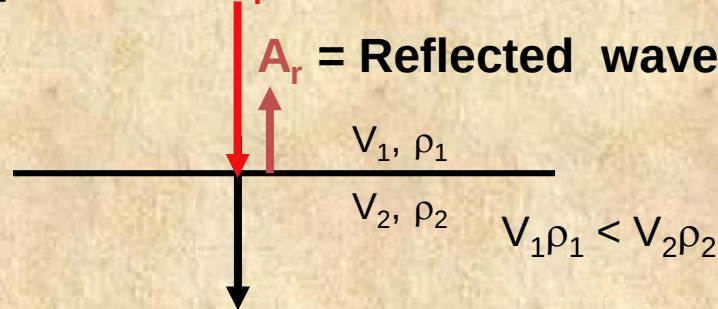
Anisotropy Example North Sea



Normal Incidence Reflection

Incident wave amplitude = A_i

A_r = Reflected wave amplitude



Transmitted wave amplitude = A_t

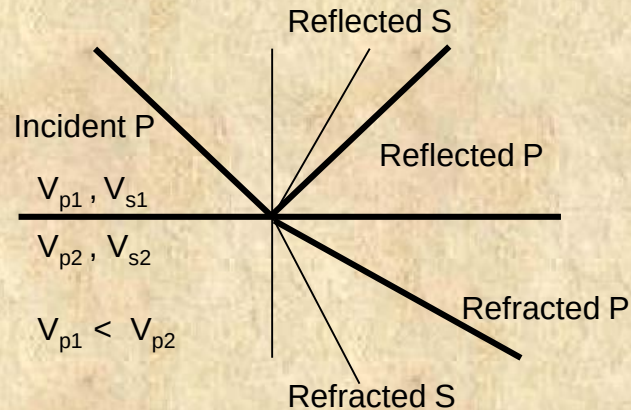
Acoustic Impedance = ρV

$$\text{Reflection Coefficient} = R = \frac{A_r}{A_0} = \left(\frac{\rho_2 V_2 - \rho_1 V_1}{\rho_2 V_2 + \rho_1 V_1} \right)$$

Transmission Coefficient = $T = 1 - R$

$$R \approx \frac{1}{2} \frac{\Delta \rho}{\rho} + \frac{1}{2} \frac{\Delta V}{V}$$

Oblique Incidence

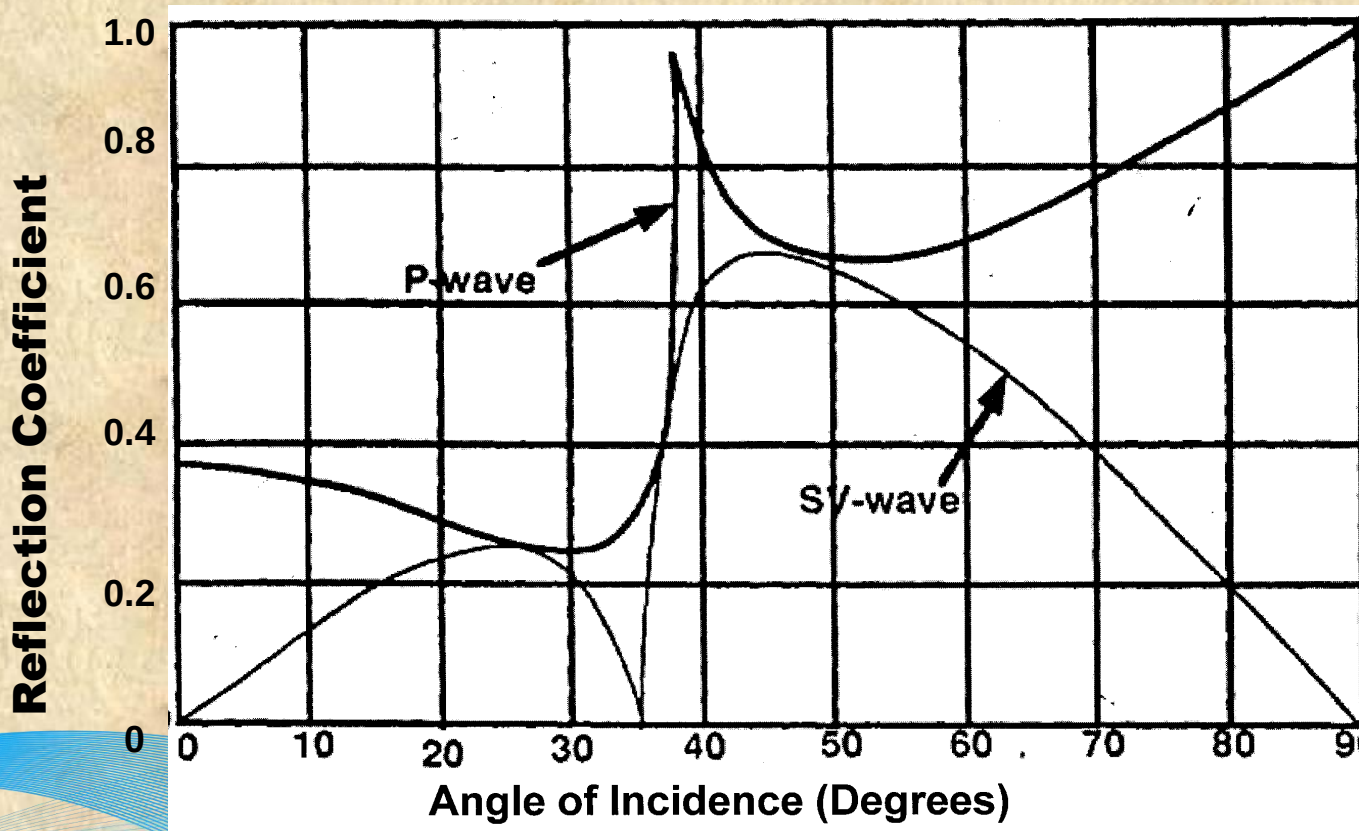


Reflected and refracted P- and S-waves generated by a P-wave obliquely incident on an interface with an acoustic impedance contrast.

Converted Waves

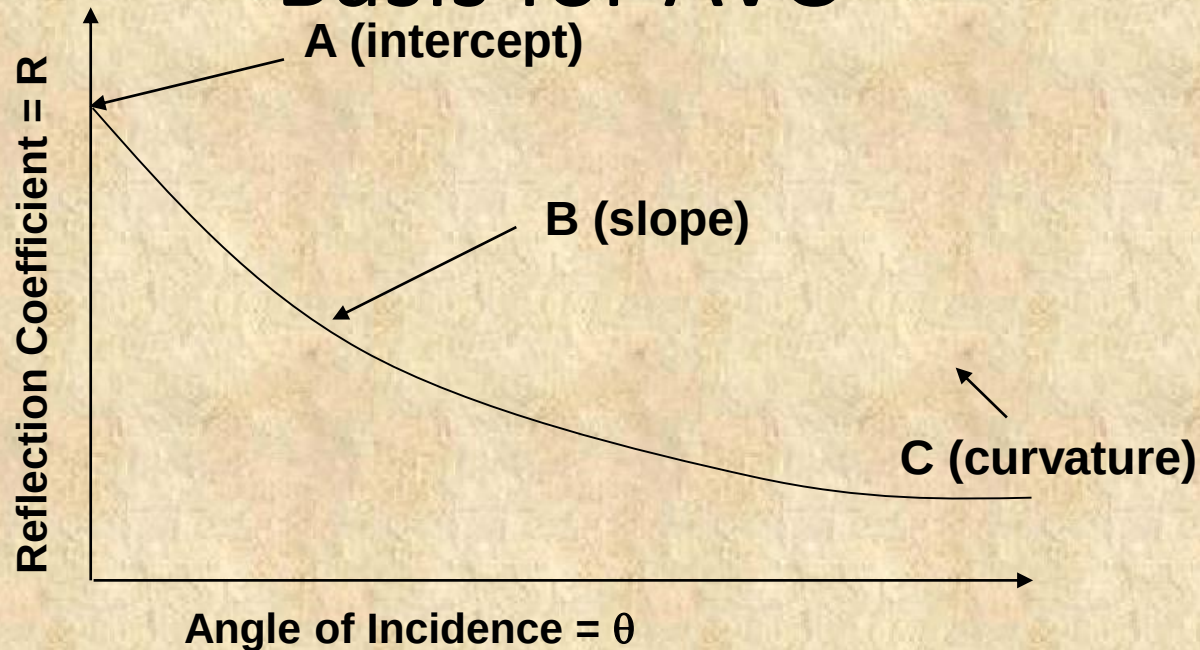
Variation of Reflection Coefficient with Angle of Incidence

$$\frac{\text{low} \longrightarrow v_{p1} v_{s1} \rho_1}{\text{High} \longrightarrow v_{p2} v_{s2} \rho_2}$$



Zoeppritz derived systems of equations that describe amplitudes of all reflected and transmitted waves

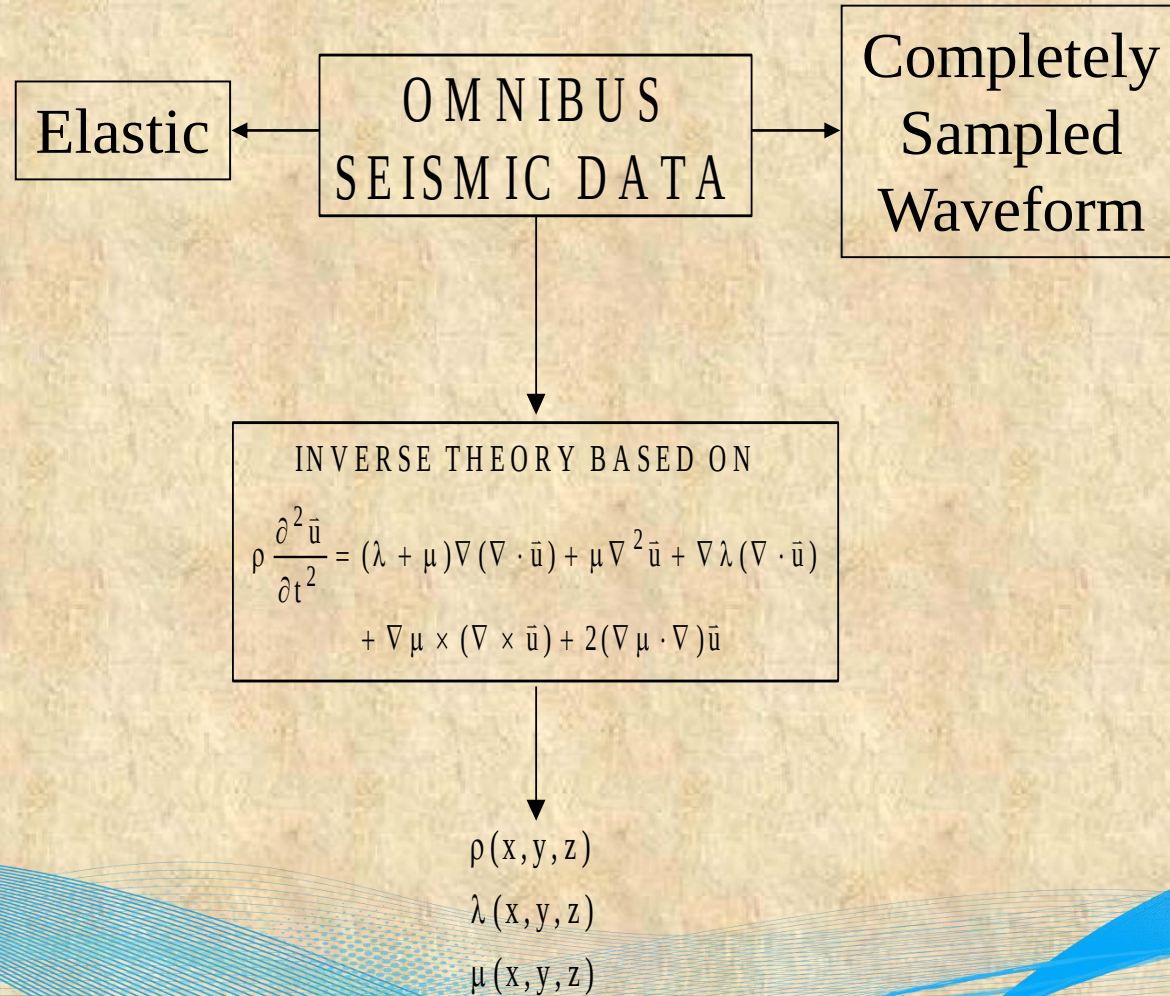
P Wave Reflection Coefficient Basis for AVO



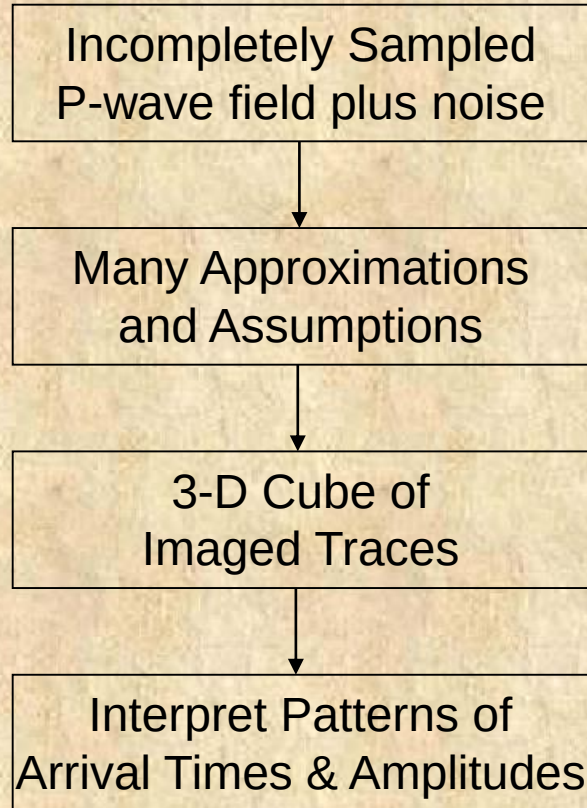
Shuey (1985) showed:

$$R(\theta) \approx A + B \sin^2 \theta + C \sin^2 \theta \tan^2 \theta + \dots$$

How Physicists Might View the Seismic Method



How We Really Do Seismic



Discussion

- List some approximations in the seismic method as we currently employ it.
- Can you rank these?
- What do we do when assumptions and approximations break down?

SUMMARY

- What we have learned about:
 - Principles of wave propagation
 - Physical basis of wave types
 - Factors that need to be considered in processing and interpretation